Rugby Intelligent Camera

1st Canto da Silva Delatorre, Tiago
Electrical and informatics department
INSA Toulouse
Toulouse, France
tcanto-da-si@insa-toulouse.fr

2nd Tassin, Nicolas
Electrical and informatics department
INSA Toulouse
City, Country
email address or ORCID

3rd Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

4th Given Name Surname
dept. name of organization (of Aff.)
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dept. name of organization (of Aff.)
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City, Country
email address or ORCID

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City, Country
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***Abstract*—Camera-based tracking systems are increasingly used in sports to enhance performance analysis and broadcast production. However, existing solutions are typically expensive and complex, making them inaccessible for amateur clubs with limited resources. This project aims to develop a cost-effective and functional camera tracking solution to, in real time, broadcast and capture the statistics of an amateur rugby match. Our proposition relies on a multi-camera setup which is controlled by a trained computer vision model, YOLO (You Only Look Once). The conducted tests show an accuracy of 80**

***Index Terms*—computer vision, rugby, cost-effective, camera tracking, YOLO**

I. Introduction

The growing availability of computer vision and deep learning technologies has transformed sports broadcasting and performance analysis. Camera-based tracking systems are now widely used in professional contexts, combining multi-camera setups with real-time image processing to follow players and the ball throughout a match. However, such systems remain complex and expensive, making them largely inaccessible to amateur clubs operating with limited resources. Several approaches have been proposed to automate sports tracking and broadcasting. YOLO-based architectures have demonstrated strong performance for real-time player and ball detection in various sports contexts(), and automated camera selection methods have been developed using spatial and temporal cues to identify the most relevant viewpoint at each moment (). Despite these advances, existing solutions have been designed primarily for professional environments and have not been adapted to the specific constraints of amateur sport. Two gaps remain unaddressed. First, no affordable, fully integrated tracking and broadcasting system has been proposed for amateur clubs. Second, ball detection in rugby presents a particularly underexplored challenge: frequent occlusions by players and an irregular ball trajectory make reliable real-time detection significantly harder than in sports such as football or basketball. To date, no study has investigated these two problems jointly. This project aimed to develop a cost-effective, autonomous system capable of capturing, processing, and broadcasting amateur rugby matches in real time, combining multi-camera recording, YOLO-based player and ball detection, and a dynamic camera selection module.

II. Related Work

Sports analysis has evolved through AI and computer vision, yet high costs exclude amateur clubs. This work establishes the theoretical foundation for a project aimed at democratizing access to automated video tracking in amateur rugby.

A. Object Detection and Capture Systems

To automate capture, the market uses panoramic stitching [1] or robotic mounts [2]. Real-time detection relies on YOLO-NAS for optimized latency [3]. Because fast-moving balls cause blur, systems integrate temporal context from adjacent frames rather than relying on single images [3], [4].

B. Trajectory Prediction and Tracking Logic

To track these detections, the Kalman Filter method predicts trajectories, maintaining continuity during occlusions. Fuzzy logic optimizations often refine these predictions to handle abrupt athletic changes, ensuring algorithms correctly link new data to existing tracks without identity switches [5].

Robust visual tracking relies on the integration of detection, temporal prediction, and dynamic state estimation rather than on perfect frame-by-frame detection. Al-Hadrusi and Sarhan [14] propose a decision-based framework combining tracking, prediction, and quality evaluation (confidence, visibility, occlusions, resolution), enabling tracking continuity during short-term observation losses. These principles are directly applicable to rugby ball tracking with a fixed camera, where the ball is small, fast-moving, and frequently occluded.

C. Spatiotemporal Feature Extraction

To enable real-time camera following, transfer learning is used within 3D Convolutional Neural Network (3D-CNN) architectures. Inflated 3D Networks (I3D) [6] is used to process spatiotemporal information. By inflating pretrained 2D filters into the temporal domain, I3D models overcome the computational bottlenecks of traditional 3D-CNNs, optimizing the balance between recognition accuracy and response latency.

Semantic Segmentation assigns pixel-level classes to isolate game elements utilizing deep models like VGG-16 with dilated convolutions[6], [7]. This approach handles class imbalance via modified cross-entropy loss, achieving high Mean IOU scores. Optical Flow quantifies motion dynamics, evolving from sparse Lucas–Kanade and dense Horn–Schunck methods[8], [9], [10] to deep architectures (FlowNet, PWC-Net) that maintain accuracy under occlusions. Action and Event Recognition combines CNN-based spatial features with LSTM-based temporal modeling to detect events in near real-time [4], [11], with robustness enhanced by multi-modal inputs like ball trajectories [12], [13].

D. Camera Localization and State Estimation

Finally, Camera Localization employs a two-stage PTZ calibration approach [14]. By utilizing deep descriptors (SuperPoint, SuperGlue) and the PTZ-IBA algorithm to minimize reprojection errors[15], [16], the system achieves high precision with efficient online processing[17].

In monocular settings, Xavier *et al.* [16] show that effective tracking can be formulated directly in the image plane using dynamic estimation corrected by visual feedback, without explicit 3D reconstruction. Complementarily, Labbé *et al.* [15] ground tracking in nonlinear observer theory, combining noisy visual measurements with dynamic models to maintain coherent state estimates (position and velocity) during occlusions or degraded observations.

E. Design Synthesis for Embedded Systems

To enable affordable rugby analysis on a Raspberry Pi, this project prioritizes computational efficiency. We conclude that a faster tracking method like YOLO-NAS is the optimal detection model due to its speed, while Kalman Filtering provides essential trajectory prediction during fast motion or occlusions. We reject complex 3D reconstruction in favor of robust 2D tracking directly on the image plane. This combination of lightweight detection and mathematical prediction ensures the system maintains real-time performance despite the hardware's strict processing limits.

III. Technical choices (explaning how we approched the problem)

A. Maintaining the Integrity of the Specifications

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IV. Our work

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A. YOLO Model

A model of Artificial Inteligence is a trained code to learn paterns based on a database[citar qualquer coisa que defina um modelo]. In between the existent models, there is the model YOLO whose algorithyme treats the image only one time (You Only Look Once).

Models like YOLO have faster responses to the video's processing, so they are usually used for real time processing, autonomus cars, monitoring, etc. The detection of plany of objects at the same time is also a caracteristc of these models, but the main reason we chose YOLO as our model was to process efficiently and with high speed.

To be able to use YOLO in python, we used the library Ultralytics[citaaaar] which makes simpler the utilisation of pre-trained models, image detection and others uses of computer vision. As so, the Figure 1 **ultralytics˙models** compares the diferentes versions of YOLO's models by latency and precision of detection. Therefore, considering the speed as main goal to track in real time during a rugby match, we chose to work with YOLOv8 version.

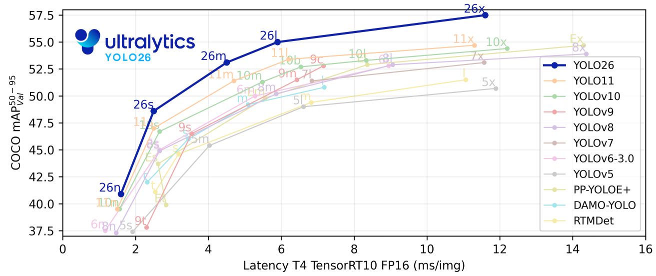

Fig. 1. Comparaison des modèles YOLO supportés par Ultralytics. Source : Ultralytics **ultralytics˙models**

Nevertheless, the Ultralytics website also presents the performance metrics of differents models of the chosen version as shown at the Figure 2 **ultralytics˙models**. To the image processing, tests were made to choose the one with good balance between accuracy and processing time.

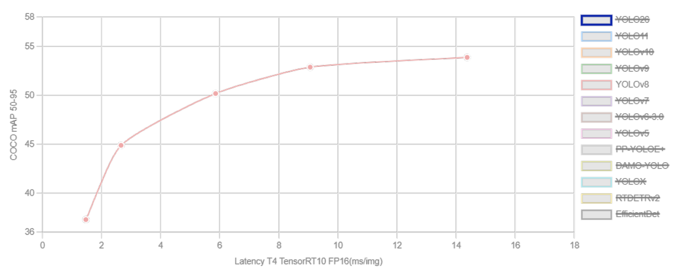

Fig. 2. Courbe de performance des différentes variantes du modèle YOLOv8. Source : Ultralytics **ultralytics˙models**

The version of the chosen model was decided by made using one frame taken from a video recorded in the real situation of a amateur match, the goal of this article, in each of the models offered by Ultralytics. The Table I, shows the results obtained in terms of processing time and quality of detection.

TABLE I
COMPARISON OF YOLOV8 MODEL VARIANTS IN THE SAME FRAME.

Model	Detected Objects	Inference	Total Time	FPS
YOLOv8n	15 persons	61.8 ms	66.0 ms	15.15
YOLOv8s	18 persons	137.5 ms	140.8 ms	7.10
YOLOv8m	21 persons	206.2 ms	208.8 ms	4.79
YOLOv8l	22 persons	616.8 ms	621.5 ms	1.61
YOLOv8x	22 persons	653.5 ms	657.2 ms	1.52

Based on these results and aiming for fast processing, we chose YOLOv8n because, even though it does not recognize as many players as the more robust models, it is more than ten times faster.

Another important conclusion is that increasing the size of the YOLOv8 model does not necessarily lead to a proportional improvement in detection performance like YOLOv8x which does not perform more efficiently on the image than YOLOv8l, even though it requires more processing time.

Although the larger models detect more players, their processing time increases significantly, which makes them less suitable for real-time video analysis. To track the focus of a rugby match, we agreed that the most important thing we should track is the ball, but in rugby the ball is small and always hidden between the players or very close to their chest, causing a big problem for the model's identification of the ball. In order to solve this problem, we added the class player; this implies that the players will also be identified by the model.

In rugby, the teams usually place themselves in lines of attack and defense, so, normally the point of interest will take place where the greatest number of players are. With both classes, we can already create a working tracking model.

The world of computer vision developers is growing and one of the biggest communities is the Roboflow Universe, which we can find a great number of models and datasets. The project Rugby Detection Computer Vision Model **b8** was the closest to our project's needs, and the pictures used in this dataset are similar to the images we captured in an INSA AS match.

After testing the model from the project **b8** in some images from the captured match (Fig. 3 to Fig. 5), positive results were achieved: the model responds well to the images, identyfing a big number of player and the ball. Some problems were: confusing the arbitrator with a player, not identifying all the players, and sometimes not seeing the ball.

Unfortunately, the model **b8** isn't accessible to download, so it wouldn't be possible to use in a real time application, because for every frame the Roboflow repository would have to be accessed on the internet, increasing the cost and delay of the image processing.

To solve the presented problem, the project's dataset **b8** was cloned and a new AI model was trained, hoping to achieve the same results as the previous. Four models were created, two with yolov8n, one with yolov8s and the last with yolov8m; they were all tested in an image not from the dataset [??? Show images of the test???].

These models aren't as good as the previous one, but they are functional for video analyses and can be used for real-time applications.

IMAGES COMPARISON

To validate our models, we tested it in a video of an INSA AS match, but the power of the available GPU wasn't enough to analyze a nineteen second video (with 1141 frames). So, we couldn't validate our models in a hole video but, analyzing in a smaller video, we can see some problems of players and ball identification, showing how the project would look like in a real-time solution.

Regarding the YOLO model we can conclude that this solution works for identification of our needs to choose the best camera but, trying to find a low-cost solution for the amateur clubs we arrived at a lack of computing power that probably will increase the cost of this approach.

To improve this project, we plan on using another YOLO model**b9** to see the statistics of the game as well, identifying tackle, penalty kick, throw-in, scrum, and other similar rugby situations that our other model can't identify. Another aimed improvement is to create a system**b9** so the model could be able to after identifying the players, separate them between

Fig. 3. Test image 1, used model from Roboflow**b8**.

Fig. 4. Test image 2, used model from Roboflow**b8**.

Fig. 5. Test image 3, used model from Roboflow**b8**.

team A and team B. The system consists of a Sigmoid loss language image pre-trained**b9** (extract color information from the player's shirt), associated with an UMAP**b9** (compact the previous information) and a Kmeans **b9** (sort both teams). These improvements would highly increase the quality of the performance analysis of the teams.

B. Video Positioning teams part

- Use either SI (MKS) or CGS as primary units. (SI units are encouraged.) English units may be used as secondary units (in parentheses). An exception would be the use of English units as identifiers in trade, such as "3.5-inch disk drive".

C. Video Acquisition and Transmission Architecture

Real-time automated tracking of amateur rugby players imposes strict constraints regarding bandwidth, latency, and hardware costs. This section details the technological choices made to ensure a fluid and synchronized video stream to the central processing unit.

D. Capture Device Choice

Several capture solutions were evaluated to meet the requirements of the multi-camera tracking project. The use of cameras on motorized platforms was initially considered for its dynamic tracking capability—which could have reduced the number of cameras needed and thus the total cost. However, extreme sensitivity to control delays and prohibitive costs led to its rejection. Similarly, solutions based on microcontrollers such as Arduino, while inexpensive, offered insufficient acquisition performance and resolution for exploitation via YOLOv8 detection models.

Ultimately, smartphones were selected as the capture units. These devices offer an optimal compromise:

- **Image Quality :** High-resolution sensors capable of providing images that can be processed by the YOLO model
- **Cost :** Utilization of existing hardware, drastically reducing the initial investment.
- **Wireless Capability :** 4G/5G cellular and Wi-Fi connection are embedded.

E. Evaluation of Streaming Protocols

To route the video streams from the smartphones to the processing computer, three main protocols were compared:

Protocol	**Infra-structure**	**Latency**	**Stability / Cost**
SRT	4G / 5G	200 ms – 500 ms	Excellent; high subscription costs
RTMP	4G/5G + remote server	2000ms	Industry standard (Youtube, Twitch); requires a server
HTTP (Local)	Wi-Fi (router)	**50 ms – 150 ms**	Excellent stability; no recurring cost

The **HTTP protocol over a local network** was selected. Unlike cellular solutions (SRT/RTMP) that depend on public network load and incur data costs, the local network allows full control, high stability, and minimizes latency.

F. Local Network Implementation and Topology

The network architecture put in place relies on the creation of a **high-performance private Wi-Fi network** (using a HUAWEI WiFi AX3-type router). This configuration makes it possible to avoid potential interference with the public networks of stadiums.

The system is structured as follows:

1) **Wireless transmission:** Two smartphones transmit their respective streams to the central access point via the HTTP protocol. Tests in real conditions validated an effective range of **80 metres**, covering a significant portion of the playing surface.
2) **Downstream link:** The router is connected to the processing computer (YOLOv8 computing unit) via a gigabit Ethernet connection. This wired link ensures that the bottleneck does not occur at the final data reception stage.
3) **Centralised processing:** The computer simultaneously receives the four streams to perform player detection and tracking.

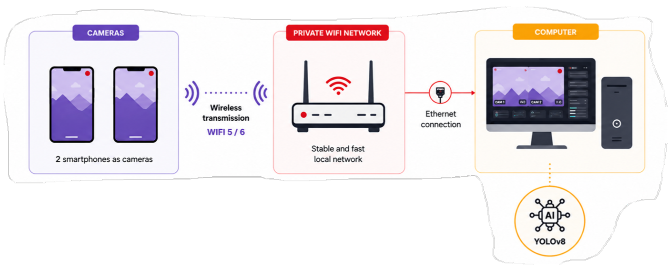

Fig. 6. Network topology: two smartphones transmit wirelessly via HTTP to a private Wi-Fi router, which forwards data via gigabit Ethernet to the processing computer.

This communication layer serves as the foundation for the Camera Score Algorithm, which must ingest these concurrent streams and dynamically prioritize the optimal feed based on the proximity of the tracked player or ball to the camera's optical axis.

G. Camera score algorithim

Please use "soft" (e.g., `\eqref{Eq}`) cross references instead

H. IHM

- The word "data" is plural, not singular.

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TABLE II
TABLE TYPE STYLES

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Fig. 7. Example of a figure caption.

Figure Labels: Use 8 point Times New Roman for Figure labels. Use words rather than symbols or abbreviations when writing Figure axis labels to avoid confusing the reader. As an example, write the quantity "Magnetization", or "Magnetization, M", not just "M". If including units in the label, present them within parentheses. Do not label axes only with units. In the example, write "Magnetization (A/m)" or "Magnetization {A[m(1)]}", not just "A/m". Do not label axes with a ratio of quantities and units. For example, write "Temperature (K)", not "Temperature/K".

Acknowledgment

The preferred spelling of the word "acknowledgment" in America is without an "e" after the "g". Avoid the stilted expression "one of us (R. B. G.) thanks ...". Instead, try "R. B. G. thanks...". Put sponsor acknowledgments in the unnumbered footnote on the first page.

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